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MILITARY SPECIFICATION

INDICATOR, VERTICAL VELOCITY, RAPID RESPONSE ACCELERATION SENSITIVE GENERAL SPECIFICATION FOR

This specification is approved for use by US Army Aviation Systems Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a rapid response acceleration-sensitive vertical velocity indicator.

1.2 Classification. Indicator shall be of the following types, as specified (see 6.2). Unless otherwise specified, Department of Army aircraft requirements are lusterless white indicators without lights.

MS25454-1 - Range 0 to 3000 feet per minute with integral lighting and lusterless white finish (Fig 1).

MS25454-2 - Range 0 to 6000 feet per minute with integral lighting and lusterless white finish (Fig 2).

MS25454-3 - Range 0 to 3000 feet per minute without integral lighting and with fluorescent luminescent finish (Fig 1).

MS25454-4 - Range 0 to 6000 feet per minute without integral lighting and with fluorescent luminescent finish (Fig 2).

MS25454-5 - Range 0 to 3000 feet per minute without integral lighting and with lusterless white finish (Fig 1).

MS25454-6 - Range 0 to 6000 feet per minute without integral lighting and with lusterless white finish (Fig 2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Aviation Systems Command, P.O. Box 209, Main Ofc, St. Louis, Missouri 63166 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors and Other Uses.
QQ-C-320	Chromium Plating (Electrodeposited).
QQ-N-290	Nickel Plating (Electrodeposited).
QQ-P-416	Plating, Cadmium (Electrodeposited).
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

MILITARY

MIL-P-116	Preservation, Method of.
MIL-D-1000	Drawings, Engineering and Associated Lists.
MIL-M-3171	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on.
MIL-N-3336	Nut, Self-Locking, Instrument Mounting.
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys.
MIL-S-6872	Soldering Process, General Specification for.
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for.

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MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys.
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges.
MIL-L-25142	Luminescent Material, Fluorescent.
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for.
MIL-M-45202	Magnesium Alloys, Anodic Treatment of.

STANDARDS

FEDERAL

FED STD No. 1	Standard for Laboratory Atmospheric Conditions for Testing.
FED STD No. 595	Colors.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-838	Lubrication of Aircraft.
MIL-STD-889	Dissimilar Metals.
MS3134	Connector, Receptacle, Electric, Solder Type, Single Hole Mounting, Push Pull Coupling.
MS25454	Indicators, Vertical Velocity, Rapid Response, Acceleration Sensitive.

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MS33558 Numerals and Letters, Aircraft Instrument Dial, Standard Form of

MS33585 Pointers, Dial, Standard Design of Aircraft Instrument.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33d Street, New York, N. Y.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.2.1.1 and 6.4). The indicator furnished under this specification shall be a product which has been tested and satisfactorily passed the first-article requirements specified in 4.4. Approval of the first articles shall not relieve the contractor of his responsibility to furnish a indicator in accordance with this specification. Any change or deviation from the approved first articles shall be subject to the prior written approval of the contracting officer.

3.2 Materials. Materials shall be as specified herein. When materials are not specified, they shall be selected in accordance with MIL-STD-143.

3.2.1 Metals. Metals shall be of a corrosion-resistant type or shall be protected to resist corrosion due to fuels, salt spray, or atmospheric conditions. Unless suitably protected against electrolytic corrosion, contact between dissimilar metals shall be avoided. Dissimilar metals are defined in MIL-STD-889.

3.2.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the instrument, except where magnetic materials are essential.

3.2.3 Fungusproof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. If any materials are

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nutrients for fungi, they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.2.4 Corrosive fumes. The materials shall not liberate deleterious fumes.

3.3 Design and construction.

3.3.1 Design. Indicator shall be designed to operate from a static pressure system and shall provide a visual indication of rates of ascent and descent from 0 to 3,000 feet per minute (fpm) as specified on Standard MS25454 for part numbers -1, -3, and -5 and from 0 to 6,000 fpm for part numbers -2, -4, and -6. A zero-setting system controlled by a slotted setting shaft shall permit the instrument to be set to indicate zero. Adjustable range of the indicator shall be not less than 400 fpm ascent to 400 fpm descent. Use of the zero-setting system shall not change the calibration of the indicator. The indicator design shall be such as to facilitate disassembly, repair or overhaul, service maintenance, and reassembly by tools and equipment normally available. Weight of indicator (excluding mounting screws and nuts, and shipping plug and cap) shall not exceed 1.75 pounds.

3.3.2 Construction. The indicator shall be constructed to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service.

3.3.3 Dimensions. Dimensions and tolerances not specified on the applicable documents (see 2.1), shall be as close as consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the instrument, they shall be held or limited accordingly.

3.3.4 Operating mechanism. The operating mechanism shall be sufficiently rugged to prevent small distortion of the case from affecting the accuracy of the instrument.

3.3.5 Gears. Gear assemblies shall be aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.3.6 Mounting screws. Mounting screws sufficient for installation of the indicator shall be furnished. The screws shall be No. 6-32NC-2 round head, nonmagnetic, and have a durable dull black oxidized or black nickel finish. Length shall be sufficient for mounting on a panel 3/16-inch thick. (See 3.9 and 5.1).

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3.3.7 Removable spring nuts. Removable spring nuts, No. 6-32NC-2, conforming to MIL-N-3336, shall be furnished to fit the mounting lugs of the indicator. (See 3.9 and 5.1.)

3.3.8 Standard parts. MS standard parts shall be used whenever they are suitable for the purpose, and shall be identified on the drawings by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins may be used provided they have suitable properties and are replaceable by standard parts without alteration. If no suitable standard part is in effect on date of invitation for bids, commercial parts may be used provided they conform to this specification.

3.3.9 Threaded parts. Screw threads 0.060-inch or larger in diameter shall be in accordance with MIL-S-7742.

3.3.10 Interchangeability of parts. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by drawing number requirements of MIL-D-1000 or as otherwise specified by the procuring activity.

3.3.11 Magnesium-alloy parts. Magnesium-alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an appropriate anodic treatment shall be used in accordance with MIL-M-45202.

3.3.12 Aluminum-alloy parts. Where practicable, aluminum-alloy parts (except the dial, if of aluminum alloy) shall be covered with an anodic film conforming to MIL-A-8625. Small holes, threads, and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.3.13 Iron and steel parts. Where practicable, iron and steel parts shall be cadmium-, or chromium-, or nickel-plated in accordance with QQ-P-416, QQ-C-320, or QQ-N-290, as applicable. Parts in a confined space in the presence of organic material shall be tin-cadmium-coated.

3.3.14 Soldering. Soldering shall be performed in accordance with MIL-S-6872.

3.3.15 Lubrication. Lubrication shall be in accordance with MIL-STD-838.

3.4 Components.

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3.4.1 Case. Outline dimensions of the case shall conform to MS25454. The case shall be airtight except for the static pressure connection. The case shall include the following parts and assemblies:

3.4.1.1 Body and mounting flange. The body and mounting flange shall be made of lightweight metallic nonmagnetic material, uniform in texture, and shall have a smooth surface. The body and mounting flange shall be finished with a durable lusterless black finish, color No. 37038 of FED-STD-595.

3.4.1.2 Bezel ring. The bezel ring shall be made of lightweight metallic nonmagnetic material, uniform in texture, and shall have a smooth surface. The bezel ring shall be finished with a durable lusterless black finish, color No. 37038 of FED-STD-595. Ring shall be held in place with screws. Provision shall be made for the replacement of the cover glass by removal of the bezel ring.

3.4.2 Dial. Dial diameter shall be a minimum of 2 3/4 inches, measured at the outside ends of the scale graduations. The pointer, numerals, at least 1/16-inch of the shortest graduations, and all other specified markings, shall be visible from any point within the frustrum of a cone whose sides form an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.4.2.1 Illumination. Integrally lighted indicators (see 1.2) shall be in accordance with MIL-L-25467.

3.4.2.2 Markings. Dial markings and background shall be in accordance with Figure 1 for part numbers -1, -3, and -5 and with Figure 2 for part numbers -2, -4, and -6 specified on MS25454. Markings shall be durable. Style and proportions of letters and numerals placed on the dial shall conform to MS33558. Fluorescent luminescent paint shall be in accordance with MIL-L-25142. Lusterless black shall be color No. 37038, and lusterless white color No. 37875, of FED-STD-595. The dial face layout and markings shall be subject to the approval of the procuring activity.

3.4.2.3 Numerals. Numerals shall distinctly indicate the graduation to which each applies. If practicable, each numeral shall be placed so that the center or mass of the numeral is on the radial line joining the appropriate graduations and the center of the dial.

3.4.3 Pointer. Except as otherwise specified, the pointer shall conform to part number -3 of MS33585. The pointer shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. It shall be firmly attached to the mechanism. Suitable stops shall be incorporated

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within the mechanism to limit the pointer movement. The pointer shall rotate to the limit marks ± 1/16-inch, for part numbers -1, -3, and -5 (Fig 1) and to 6,000 fpm graduations, ± 1/16-inch, for part numbers -2, -4, and -6 (Fig 2), specified on MS25454. The pointer shall be marked in accordance with Figures 1 and 2 and the provisions of 3.4.2.2. The pointer tip width should be the same as the 100 ft index marks (.030 width) and the pointer length should be such that the tip extends up to but not overlap the 100 ft graduation marks with a tolerance of ± .010.

3.4.4 Cover glass. Cover glass shall be in accordance with type II, quality AA of DD-G-451. Distance between the dial and inside of the cover glass shall be the minimum practicable and shall not exceed 0.250-inch. Anti-reflective coatings on all glass surfaces are required in accordance with MIL-C-14806.

3.4.5 Electrical receptacle. The electrical receptacle used for the integral lighting (see 3.4.2.1) shall conform to part number 3P of MS3134. The receptacle shall be located on the back of the case. The polarizing key of the receptacle shall be at the top center of the receptacle shell.

3.5 Performance. The indicator shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.6 Finish. Protective coatings and finishes shall not crack, chip, or scale during normal service life or due to extremes of atmospheric conditions.

3.7 Identification of product. Indicator shall be identified in accordance with MIL-STD-130.

3.8 Manufacturer's part number. The manufacturer's part number shall be identical with the manufacturer's engineering production drawing number, including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.9 Workmanship. The indicator, including all parts, shall be constructed and finished to produce an instrument free from any defect which could affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding, brazing, painting, machine-screw assemblies, and freedom of parts from burrs and sharp edges. Prior to final assembly, the instrument shall be thoroughly cleaned of loose, spattered, or excess solder, resin flash that may crumble, metal chips, and other foreign matter.

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3.10 Pressure port plug and connector cap. A plug, for the static pressure port, and a cap for the connector, in indicators with integral lighting, shall be installed. The plug shall allow the indicator to breathe, precluding damage from differentiating pressures (4.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Inspection and testing of indicators shall be classified as (1) first article and (2) acceptance.

4.2.1 First article. First article shall consist of 3 indicators of the manufacturer's part number. All three samples shall be subjected to first article tests (see 4.2.1.1). A test report prepared in accordance with MIL-STD-831 shall certify conformance with first article tests and shall be submitted to the contracting officer for approval before delivery of production indicators.

4.2.1.1 First article shall be tested as follows:

(a) Individual tests (see 4.2.2.1).
(b) Sampling tests (see 4.2.2.2.3).
(c) Vibration endurance (see 4.4.14).
(d) Magnetic effect (see 4.4.15).
(e) High altitude, low temperature (see 4.4.16).
(f) High temperature exposure (see 4.4.17).
(g) Humidity (see 4.4.18).
(h) Fungus (see 4.4.19).
(i) Salt spray (see 4.4.20).
(j) Illumination (for Dash 1 and Dash 2 of Standard MS25454) (see 4.4.7.1).
(k) Mounting lugs (see 4.4.21).

4.2.2 Acceptance. Acceptance tests shall consist of the individual and sampling tests.

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4.2.2.1 Individual. Each indicator shall be submitted to the individual tests speficied below. Any indicator failing to pass any of these tests shall be rejected.

(a) Examination of product (see 4.4.1).
(b) Zero setting (see 4.4.2).
(c) Leakage (see 4.4.3).
(d) Scale error at room temperature (see 4.4.4).
(e) Pointer lag (see 4.4.5).
(f) Illumination (for Dash 1 and Dash 2 only of MS25454) (see 4.4.7.2).
(g) Position error (see 4.4.8).

4.2.2.2 Sampling tests. The test samples selected for sampling tests shall first have passed the individual tests. The test samples which have been subjected to sampling tests shall not be delivered on contract until they have been refurbished and resubjected and passed all the individual tests.

4.2.2.2.1 Sample test sample selection. Sampling test samples shall be selected at random in accordance with the following schedule:

QUANTITY OF INDICATORS OFFERED FOR ACCEPTANCE	QUANTITY TO BE SELECTED FOR TESTING
First 15	1
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

4.2.2.2.2 Sampling test defects. When a defect occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of the first fifteen and proceed as indicated.

4.2.2.2.3 Sampling tests. Sampling tests shall include:

(a) Illumination (for Dash 1 and Dash 2 only of MS25454) (see 4.4.7.3).

(b) High temperature (see 4.4.9).

(c) Low-temperature zero error (see 4.4.10).

(d) Scale error during temperature change (see 4.4.10.1).

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(e) Low-temperature scale error (see 4.4.10.2).

(f) Zero error during temperature change (see 4.4.11).

(g) Vibration error (see 4.4.12).

(h) Over pressure (see 4.4.13).

(i) Weight (see 4.4.22).

4.3 Test conditions.

4.3.1 Atmospheric. Unless otherwise specified herein, all tests shall be made under atmospheric conditions specified in FED-STD-1. Ambient conditions stated in MIL-STD-810.

4.3.2 Tapping. Unless otherwise specified, the indicator shall be tapped before a test reading is taken. An electrical vibrator set at 30 to 60 cycles per second (cps) with a maximum amplitude of 0.002-inch when attached to the test item may be used.

4.3.3 Vibration stand. Whenever a vibration stand is specified (see 4.4.12), it shall be a device which will vibrate at any desired frequency between 2.5 and 50 cps, constantly varying the amplitude from .25-inch at 2.5 cps to .020-inch at 50 cps.

4.3.4 Position of indicator. Unless otherwise specified herein, the indicator shall be tested in its normal operating position with the dial vertical.

4.4 Test methods.

4.4.1. Examination of product. The indicator shall be inspected to determine compliance with this specification with respect to materials, workmanship, and marking.

4.4.2 Zero setting. The adjusting screw shall be moved throughout its range in both the UP and DOWN directions. The adjustable range of pointer movement shall be not less than 400 feet per minute (fpm) UP, to not less than 400 fpm DOWN.

4.4.3 Leakage. The static pressure connection of the indicator shall be suitably connected to a mercury manometer and sources of vacuum and pressure. A vacuum of 15 inches Hg shall be applied to the static pressure connection at a rate not exceeding 20,000 fpm. With the source closed off, the difference between the mercury levels in the manometer during a period of 1 minute shall not exceed 0.05-inch Hg. The same procedure shall then

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be repeated with a pressure of 10 inches Hg. The same tolerance shall apply.

4.4.4 Scale error at room temperature. The indicator, with the pointer set to zero, shall be placed in an altitude chamber. A mercury barometer shall be connected to the chamber. Pressure shall be reduced at a rate such that the pointer indicates 500 fpm. While the pointer indicates 500 fpm, time required for the mercury barometer to change from a pressure corresponding to 1,000 feet altitude to that of an altitude of 1,500 feet shall be determined by a stop watch. Pressure shall then be increased at such a rate that the pointer will indicate a descent of 500 fpm and the time required for the barometer to change from 1,500 feet altitude to 1,000 feet altitude determined. This procedure shall be continued for the test rates and altitudes shown in Table I. Scale error shall not exceed the tolerances listed in Table I.

4.4.4.1 Scale error determination. When testing several instruments in one chamber at the same time, one instrument may be selected to indicate the test rate. This test rate shall be maintained at a constant value while test readings of all instruments are observed and noted, and while the time and altitude interval on the barometer are noted. The computed rate shall be obtained. Scale error shall be determined.

4.4.5 Pointer lag. The static pressure connection of the indicator shall be suitably connected to a water manometer and sources of vacuum and pressure. A vacuum shall be carefully applied to the static pressure connection (the indicator will show a rate of ascent) sufficient to cause the pointer to indicate a descent of more than 3,000 fpm, after the pointer has passed through the zero position when this vacuum is instantaneously released by opening the static pressure tubing to atmosphere. The indicator shall be tapped during the return of the pointer to zero. A stop watch shall be started at the instant the pointer passes the 3,000 fpm graduation and stopped when the pointer passes the 300 fpm graduation. The time shall be recorded. This procedure shall be repeated, except that a slight pressure shall be carefully applied to the static pressure connection and the time determined for the indication to change from 3,000 fpm climb to 300 fpm climb. The time recorded shall be not less than 7 seconds nor more than 12 seconds. During this test and at any time after vacuum (or pressure) is released, the pointer shall move smoothly without excessive friction.

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TABLE I. Scale Error at Room Temperature

Standard Altitude Interval	Test Rate	Tolerance Scale Error Ascent or Descent	Type of Indicator to Which Applied	
Feet	Feet per Minute	Feet per Minute	MS25454-1, -3, -5	MS25454-2*, -4, -6
1,000 to 1,500	500	100	(*)	(*)
1,000 to 2,000	1,000	200	(*)	(*)
2,000 to 4,000	1,000	200	(*)	
	1,500	200	(*)	
	2,000	300	(*)	(*)
	3,000	300	(*)	(*)
	4,000	400	.....	(*)
	5,000	500	.....	(*)
15,000 to 17,000	1,000	200	(*)	(*)
	2,000	300	.....	(*)
	3,000	300	(*)	
	4,000	400	.....	(*)
28,000 to 30,000	1,000	200	(*)	
	2,000	300	.....	(*)
	3,000	300	(*)	
	4,000	400	.....	(*)

*Indicator Applicability.

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4.4.6 Acceleration response. With the pointer zeroed and the dial in a vertical plane, the indicator shall be quickly turned over to an inverted position. The pointer shall rapidly indicate a maximum descent reading of 2,300 ± 500 fpm. It shall hold the maximum reading and suddenly break and return moderately fast to zero. The time from ROLL OVER to BREAK shall be 3 ± 1 seconds, and the time from BREAK to the 500-fpm graduation shall be 3.7 ± 1 seconds. After the instrument has settled at zero, the indicator shall be rapidly rolled over to its normal position. The pointer shall deflect up to a maximum of 700 ± 300 fpm. The time from ROLL OVER until the pointer returns to the 200-fpm graduation shall be 20 ± 6 seconds. During this test, vibration at an amplitude of 0.002 inch and a frequency of approximately 30 to 60 Hz shall be applied to the indicator.

4.4.7 Illumination. Illumination tests are applicable for Dash 1 and Dash 2 only of MS25454.

4.4.7.1 First articles (4.2.1.1). Indicator shall be tested in accordance with the test methods specified in MIL-L-25467.

4.4.7.2 Individual tests (4.2.2.1). Indicator shall be tested in accordance with the operation test specified in MIL-L-25467.

4.4.7.3 Sampling (4.2.2.2.3). Indicator shall be tested in accordance with the light test specified in MIL-L-25467.

4.4.8 Position error. The indicator shall be held in the normal operating position and tapped, and a reading taken. Instrument readings for the following positions shall be obtained and recorded.

(a) Rotated from normal operating position through 90 degrees to the rear (face up).

(b) Rotated from normal operating position through 90 degrees to the front (face down).

(c) Rotated from normal operating position through 90 degrees to the right (right side down).

(d) Rotated from normal operating position through 90 degrees to the left (left side down).

The maximum deviation of the four latter readings from the original reading shall not exceed 50 fpm. The readings shall be taken after the indicator has been tapped and the pointer has remained in a steady position for not less than 5 seconds.

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4.4.9 High temperature. The indicator shall be subjected to a temperature of 71° C. for a period of 3 hours. It shall then be allowed to cool to room temperature, after which it shall be subjected to the tests for scale error at room temperature (4.4.4) and leakage (4.4.3). The indicator shall meet the requirements specified for scale error at room temperature, except that the intervals and rates listed in Table II shall be used. During and after this test, the setting system shall operate satisfactorily.

TABLE II. Scale Error During Temperature Tests

Standard Altitude Interval	Test Rate	Tolerance Scale Error Ascent or Descent	Type of Indicator To Which Applied	
Feet	Feet per Minute	Feet per Minute	MS25454 -1,-3,-5	MS25454 -2,-4,-6
2,000 to 4,000	1,000	200	(*)	
	2,000	300	(*)	(*)
	4,000	400	.....	(*)
15,000 to 17,000	1,000	200	(*)	
	2,000	300	(*)	(*)
	4,000	400	.....	(*)
28,000 to 30,000	1,000	200	(*)	
	2,000	300	(*)	(*)
	4,000	400	.....	(*)

*Indicator Applicability.

4.4.10 Low-temperature zero error. Before starting the low-temperature zero error test, the pointer shall be set to zero at room temperature. The indicator shall then be placed in a test chamber and cooled to approximately -54° C. within any length of time between 15 minutes and 5 hours, and allowed to remain at the temperature for a period of not less than 3 hours. With the test chamber at atmospheric pressure, a low-temperature zero error (during the cooling period and during the 3-hour period at -54° C), shall not exceed 150 fpm.

4.4.10.1 Scale error during temperature change. At 1-hour intervals during the cooling period and during the 3-hour period at -54° C., indicated rates of the indicator corresponding to the test rates of

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The difference between the indication of the pointer at the beginning and at the end of the vibration cycling period shall not exceed 50 fpm. In addition, no looseness in the mechanism nor damage to any part of this test, the indicator shall be subjected to, and meet, the individual tests (see 4.2.2.1).

4.4.15 Magnetic effect. The indicator shall be held in various positions with its nearest part 5 inches from, and magnetically east or west, the center of a short bar magnet compass. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. This test shall be made with the integral lighting off and repeated with the lighting on. Deflection of the compass shall not exceed 1 degree.

4.4.16 High altitude - low temperature. The indicator shall be placed in an altitude chamber maintained at -65° C. and pressure reduced to simulate an altitude condition of 50,000 feet (3.436 inches Hg. absolute pressure) for a period of 4 hours. The indicator shall then be maintained at -65° C., at atmospheric pressure, for an additional period of 44 hours. After returning to room temperature, the indicator shall be carefully examined for any evidence of damage, and shall be subjected to, and pass, the individual tests (see 4.2.2.1) (MIL-STD-810, Method 504, Procedure I).

4.4.17 High-temperature exposure. The indicator shall be subjected to 71°, ± 2° C., for a period of 24 hours. At the end of the 24-hour period and while still at high-temperature, the indicator shall operate without indication of sticking or erratic action. After the chamber has been returned to room temperature and the indicator has saturated at room temperature for a minimum period of 4 hours, it shall be subjected to, and pass, the individual tests (see 4.2.2.1). There shall be no damage which could adversely affect subsequent operation. (MIL-STD-810, Method 501, Procedure I).

4.4.18 Humidity. The indicator shall be subjected to the humidity test, Procedure I of MIL-STD-810, Method 507. To simulate installed conditions, the external connections to the indicator shall be connected with 10 feet of 1/4-inch copper or aluminum tubing coiled so the moisture can drain out of the open end. Within 1 hour after the 360-hour period, the indicator shall be subjected to, and pass, the scale error at room temperature test (see 4.4.4).

4.4.19 Fungus. The indicator shall be subjected to the fungus resistance test. Procedure I of MIL-STD-810, Method 508, except the duration of the test shall be 14 days.

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2,000 fpm climb and descent, in the standard altitude interval between 2,000 and 4,000 feet, shall be obtained. These indicated rates, corrected for the corresponding zero errors, shall not exceed the values listed in Table II.

4.4.10.2 Low-temperature scale error. After a temperature of -54° C. has been maintained within the chamber for 3 hours, the indicator shall be subjected to the test specified for scale error at room temperature (see 4.4.4) except that intervals and rates specified in Table II shall be used. The indicated rates, corrected for the low-temperature zero error, at the end of the 3-hour period at -54° C., shall not exceed the values specified in Table II. During and after this test, the setting system shall operate satisfactorily.

4.4.11 Zero error during temperature change. After the low-temperature scale error test has been conducted, the indicator shall be allowed to return to room temperature at atmospheric pressure by immediate removal from a -54° C. to room temperature. While temperature of the indicator is increasing, the indication shall be observed for a period of 1/2-hour, at 1-minute intervals for the first 5 minutes and at 5-minute intervals thereafter. During the first 5-minute interval, the maximum reading shall not exceed 300 fpm; thereafter, the maximum reading shall not exceed 200 fpm.

4.4.12 Vibration error. The indicator shall be mounted on a vibration stand in its normal operating position with the dial vertical. For this test, the pointer shall be set to zero. No pressure or vacuum shall be applied during the test. The indicator shall be subjected to vibration with an amplitude between 0.003- and 0.005-inch at frequencies of 5 to 50 cps in order to determine that natural frequency of the indicator does not occur in this frequency range. While the indicator is being vibrated, maximum range of pointer oscillation at any time shall not exceed 50 fpm. Pointer variation from the zero position shall not exceed 50 fpm.

4.4.13 Overpressure. With the indicator in an altitude test chamber after the pointer has been set to zero, the pressure shall be decreased at the rate of 15,000 fpm for MS25454-1 and -3 & -5 indicators, or 30,000 fpm for MS25454-2 and -4 & -6 indicators, until pressure corresponding to 50,000 feet is reached. The pressure shall then be increased at a rate of 30,000 fpm for MS25454-1 and -3 & -5 indicators and 60,000 fpm for MS25454-2 and -4 & -6 indicators, until room pressure is reached. Thereafter, the indicators shall be subjected to, and pass the individual tests specified in 4.2.2.1.

4.4.14 Vibration endurance. The indicator shall be subjected to the vibration test of MIL-STD-810, Method 514.2, Procedure I, Part I, Curve M, using the time schedule of Table 514.2-III.

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4.4.20 Salt spray. The indicator shall be subjected to the salt spray test. Procedure I of MIL-STD-810, Method 509, for a period of 50 hours. Installed conditions of the humidity test shall apply. At the end of the 50-hour period, the indicator shall be subjected to, and pass, the individual tests (see 4.2.2.1).

4.4.21 Mounting lugs. The indicator case, with mechanism removed, shall be mounted face downward on the movable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting nut and attached to a pull strap. A load of 175 pounds shall be applied along the axis toward the front of the case for a period of 1 minute. There shall be no breakdown.

4.4.22 Weight. Indicator shall not exceed 1.75 pounds (excluding the mounting screws, nuts, and shipping plug and cap).

4.5 Packaging, packing and marking. Preparation for delivery shall be inspected for conformance with Section 5 and paragraph 3.10 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Indicators shall be packaged level A, B, or C, as specified (see 6.2). Installation instructions, mounting screws (see 3.3.6), and removable spring nuts (see 3.3.7) necessary for installation shall be placed in an envelope and packaged with each indicator. The envelope shall be marked as follows: IMPORTANT - THIS ENVELOPE CONTAINS INSTALLATION INSTRUCTIONS, MOUNTING SCREWS, AND SPRING NUTS.

5.1.1 Level A. Each indicator shall be packaged Method III of MIL-P-116, wrapped and cushioned as required, in a snug-fitting fiberboard container conforming to W6s or W6c of PPP-B-636.

5.1.2 Level B. Each indicator shall be packaged as for level A (see 5.1.1), except that the fiberboard container may be domestic class.

5.1.3 Level C. Indicators shall be packaged in a manner to prevent corrosion or damage during shipment from the supply source to the first destination for immediate use.

5.2 Packing. Indicators, packaged in accordance with 5.1, shall be packed level A, B, or C, as specified (see 6.2).

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5.2.1 Level A. Indicators shall be packed in close-fitting wood-cleated plywood boxes conforming to Grade A, Style I or J of PPP-B-601; or in nailed wood boxes conforming to Class 2 of PPP-B-621. Gross weight of the pack shall not exceed 200 pounds.

5.2.2 Level B. Indicators shall be packed as for level A (see 5.2.1) except that fiberboard boxes conforming to V3s, V4s, or V3c of PPP-B-636 may be used.

5.2.3 Level C. Indicators shall be packed in a manner to assure carrier acceptance and safe delivery to destination.

5.3 Marking. In addition to any special requirements of the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary markings. Whenever practicable, the following precautionary markings shall appear on two opposite sides of each interior package and shipping container: FRAGILE, DELICATE INSTRUMENT(s), HANDLE WITH CARE.

6. NOTES

6.1 Intended use. The acceleration-sensitive rapid response vertical velocity indicator is intended for use on aircraft to indicate the vertical component of the aircraft's speed based on the rates of change of atmospheric pressure and the effects of vertical acceleration.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type of indicator required (see 1.2).

(c) Level of packaging and of packing required (see 5.1 and 5.2).

(d) Whether special markings are required (see 5.3).

(e) When a first article is required for inspection and approval (see 3.1, 4.2.1, and 6.4).

6.3 Definitions

6.3.1 Amplitude. Extent of motion as measured from one extreme to the opposite extreme.

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6.3.2 Test reading. Indication of the instrument during the test.

6.3.3 Test rate. The rate at which the instrument is to be tested as specified in Column 2 of Tables I and II.

6.3.4 Computed rate. The rate computed by dividing the altitude interval (2,000 feet as measured by a barometer) by the time in seconds (required for the change in altitude) and multiplying by 60.

$$\text{Computed rate} = \frac{\text{Altitude interval} \times 60}{\text{Time in seconds}}$$

6.3.5 Computed rate correction. The algebraic difference between the test rate and the computed rate is "computed rate correction."

Example 1.

Test rate	+ 2,000
Computed rate	+ 1,925
Computed rate correction	+ 75

Example 2.

Test rate	+ 2,000
Computed rate	+ 2,025
Computed rate correction	- 25

6.3.6 Indicated rate. The reading of the instrument as computed herein to correspond to a "test rate." The indicated rate of the instrument should be computed by adding algebraically the test reading of the instrument and the computed rate correction.

6.3.7 Scale error. The difference between the "test rate" and the "indicated rate" is the "scale error."

6.4 First article inspection. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a sample selected from the first 3 production items, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of 3 units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or

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test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement. It should be understood that the product supplied under contract or order will be identical to the corresponding first article in design, construction, quality, material, workmanship, and method of manufacture. Deviation from the standards of the first article may be made only by the procuring activity. Evidence of unauthorized change will constitute cause for rejection.

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Army - AV

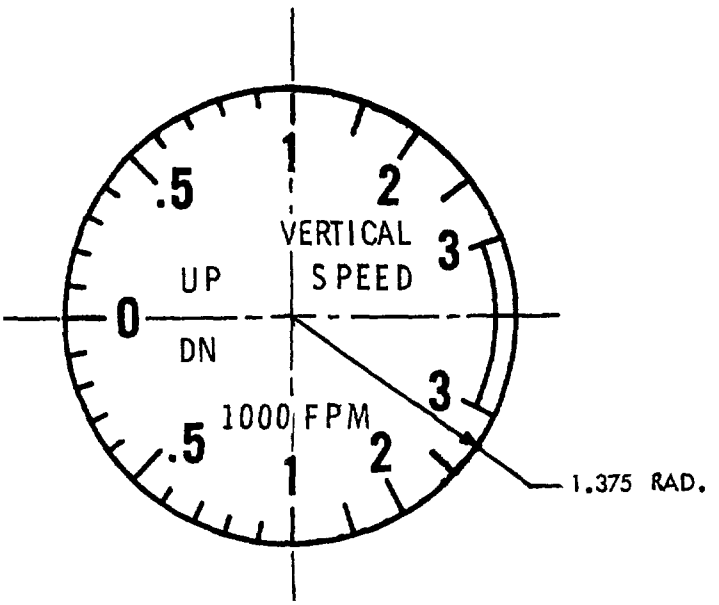
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in.	mm
.10	2.54
.15	3.81
.16	4.064
.20	5.08
.22	5.588
.25	6.35
.30	7.62
.0125	0.3175
.020	0.508
.025	0.635
.030	0.762
.035	0.889
.0375	0.9525
.050	1.27





MS33585-3

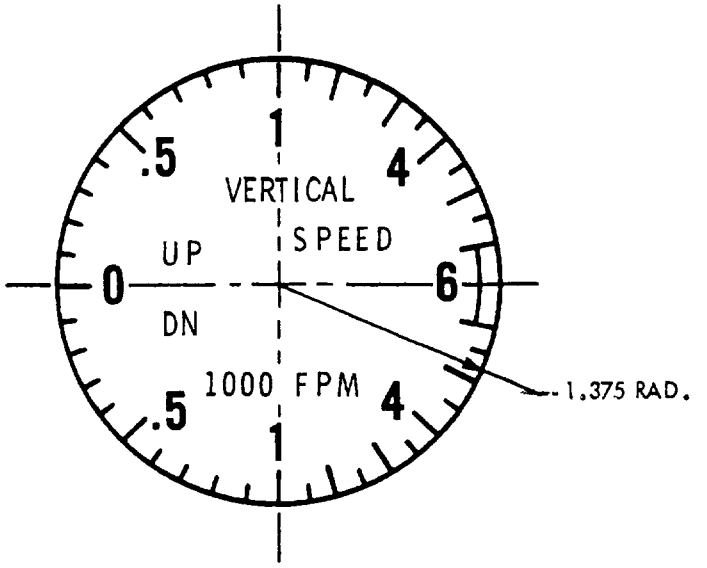
0 TO 3,000

MARKING	HEIGHT OR LENGTH	WIDTH OF LINE	FINISH
ZERO GRADUATION	.25	.050	LUSTERLESS WHITE (SEE 1.2)
1,000 FT GRADUATIONS	.22	.035	
.5 INDEX GRADUATIONS	.22	.035	
ALL OTHER 500 FT GRADUATIONS	.16	.030	
100 FT GRADUATIONS	.16	.030	
NUMERALS .5	.3	.0375	
NUMERALS 0,1,2,3	.3	.0375	
LETTERING "VERTICAL SPEED"	.15	.020	
ARC CONNECTING 3000 FT. MARKS	—	.030	
LETTERING "UP AND DOWN"	.20	.025	
LETTER "1,000 FPM"	.10	.0125	
UNSHADED PART OF POINTER	—	—	LUSTERLESS BLACK
BACKGROUND OF DIAL	—	—	

FIGURE 1. DIAL MARKINGS FOR TYPE MS25454-1, -3 AND -5 INDICATORS

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in.	mm
.10	2.54
.15	3.81
.16	4.064
.20	5 08
22	5.588
.25	6.35
30	7.62
.0125	0.3175
020	0.508
025	0.635
.030	0 762
.035	0.889
.0375	0 9525
.050	1.27





MS33585-3

0 TO 6,000

MARKING	HEIGHT OR LENGTH	WIDTH OF LINE	FINISH
ZERO GRADUATION	.25	.050	LUSTERLESS WHITE (SEE 1.2)
1,000 FT GRADUATIONS	.22	.035	
.5 INDEX GRADUATIONS	.22	.035	
ALL OTHER 500 FT GRADUATIONS	.16	.030	
100 FT GRADUATIONS	.16	.030	
ARC CONNECTING 6,000 FT MARKS	—	.030	
NUMERALS .5	.3	.0375	
NUMERALS 0,1,2,4,6	.3	.0375	
LETTERING "VERTICAL SPEED "	.15	.020	
SHADED PART OF POINTER	—	—	
LETTERING"UP AND DOWN"	.20	.025	
LETTER "1,000 FPM"	.10	.0125	
UNSHADED PART OF POINTER	—	—	LUSTERLESS BLACK
BACKGROUND OF DIAL	—	—	

FIGURE 2. DIAL MARKINGS FOR TYPE MS25454-2, -4 AND -6 INDICATORS

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